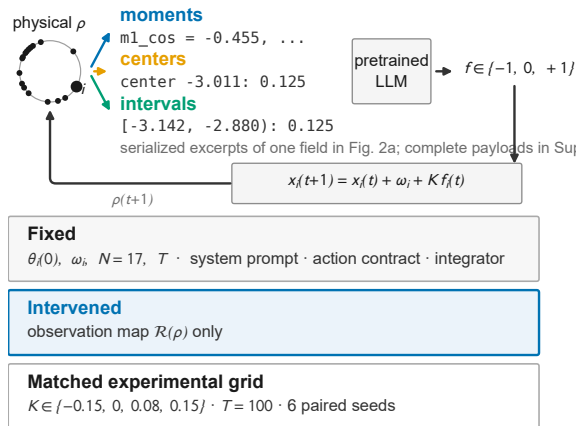
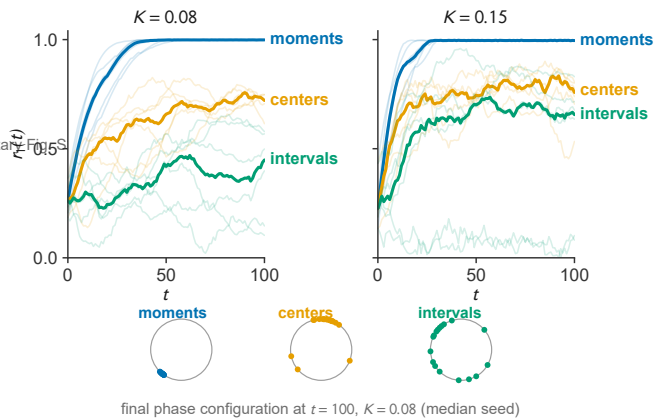
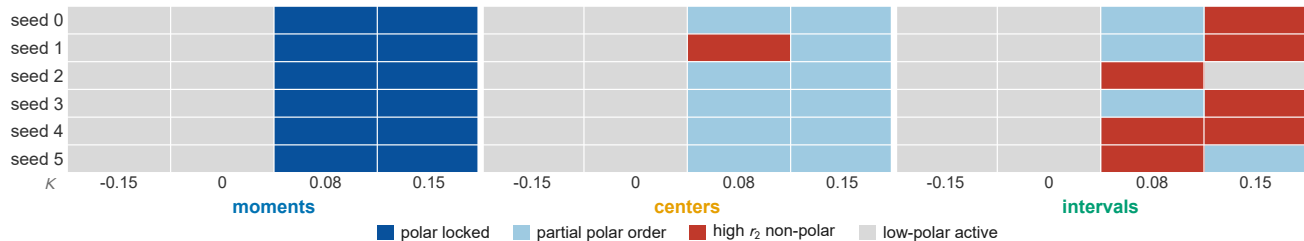
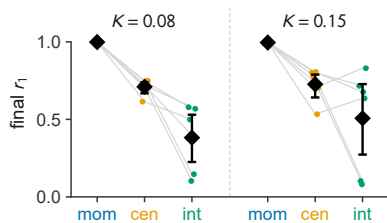
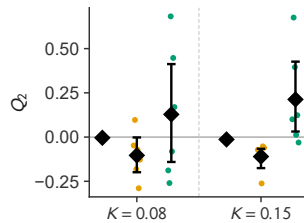
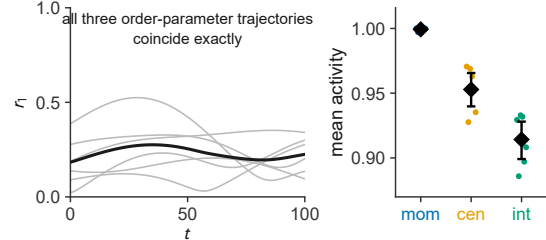


**a** Matched observation-map intervention on the same physical state**b** Collective dynamics at positive coupling**c** All-seed phenotype matrix**d** Paired final  $r_1$ **e** Final  $Q_2 = r_2 - r_1$ **f**  $K=0$  negative control

the collective is identical; the microscopic operator is not